

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. MTB-601: Analysis-II

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each unless specified otherwise.)

- (a) If $f(z)$ and $\overline{f(z)}$ are both analytic functions defined in a domain D , show that $f(z)$ is constant. [2]
- (b) Find the relation among the real constants a, b, c so that the function $f(z) = (x + ay) + i(bx + cy)$ is entire. [3]
- (c) Find the image of the line $y = 2x$ under the mapping $w = (1 + i)z + 2$. [2]
- (d) Find the bilinear transformation which maps the points $z = i, 2, -2$ into $w = i, 1, -1$ respectively. [3]
- (e) Let (X, d) be a metric space and let C be a nonempty subset of X . Let $d_C : X \rightarrow \mathbb{R}$ be defined by:

$$d_C(x) = \inf_{y \in C} d(x, y), \quad \text{for all } x \in X$$

Show that:

$$|d_C(x) - d_C(y)| \leq d(x, y), \quad \text{for all } x, y \in X$$

[3]

- (f) Let $X = \{\frac{1}{n} : n \in \mathbb{N}\} \cup \{0\}$ be equipped with the usual metric. Let $A = \{\frac{1}{2n} : n \in \mathbb{N}\}$ and $B = \{\frac{1}{2n+1} : n \in \mathbb{N}\}$. Find the distance $d(A, B)$. [2]
- (g) Let Y be a subset of a metric space (X, d) and let C be a subset of Y . Show that $x \in Y$ is a limit point of C in Y if and only if x is a limit point of C in X . [4]
- (h) Let $\mathbb{Q}$ be the set of all rational numbers. Show that $\mathbb{Q}$ is not complete under the usual metric. [3]

Question 2.

- (a) Show that the cross-ratio is invariant under a Möbius (bilinear) transformation. ... [6]
- (b) Derive the Cauchy-Riemann equations in Cartesian form for the analyticity of the function $f(z) = u(x, y) + iv(x, y)$. [6]

OR

- (a) Show that a bilinear transformation maps a circle or a straight line of the z -plane onto a circle or a straight line of the w -plane. [6]

- (b) Discuss the image of the open upper semi-unit disc:

$$\Omega = \{z \in \mathbb{C} : |z| < 1, \operatorname{Im}(z) > 0\}$$

under the transformation:

$$w_a(z) = \frac{z - a}{1 - \bar{a}z}$$

where a is a complex number such that $|a| < 1$. [6]

Question 3.

- (a) Let d and d' be metrics on a nonempty set X . Show that:

- (i) $d + d'$ is a metric on X ;
- (ii) $\max\{d, d'\}$ is a metric on X .

[6]

- (b) Let G be a subset of a metric space (X, d) . Show that the following statements are equivalent: [6]

- (i) G is open.
- (ii) G is the union of open balls in X .

Question 4.

- (a) Let Y be a subspace of a metric space (X, d) and U a subset of Y . Show that U is open in (Y, d_Y) if and only if there exists an open set G in X such that $U = Y \cap G$. [6]

- (b) Let C be a subset of a metric space (X, d) . If the closure $\bar{C} = C \cup C'$ (where C' is the set of all limit points of C), show that: [6]

- (i) $\bar{C}$ is closed;
- (ii) $\bar{C}$ is the smallest closed subset of X containing C .

Question 5.

- (a) Let C be a nonempty subset of a metric space (X, d) . Show that:

$$\operatorname{diam}(C) = \operatorname{diam}(\bar{C})$$

[6]

- (b) Show that the following metrics defined on $\mathbb{R}^n$ (where $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$) are equivalent: [6]

- (i) $d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$
- (ii) $d_1(x, y) = \sum_{i=1}^n |x_i - y_i|$
- (iii) $d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

Question 6.

- (a) Let $X = \ell_\infty$ be the set of all bounded sequences of real numbers, and let d be defined by:

$$d(x, y) = \sup_{n \in \mathbb{N}} |x_n - y_n|, \quad \text{for } x, y \in X$$

Show that (X, d) is a complete metric space. [6]

- (b) State and prove Cantor's Intersection Theorem. [6]

Question 7.

- (a) Let (X, d) and (Y, ρ) be metric spaces, and $f : X \rightarrow Y$ be a function. Show that the following statements are equivalent: [6]

(i) f is continuous at $x_0 \in X$.

(ii) $\{f(x_n)\}_{n=1}^\infty$ converges to $f(x_0)$ in Y whenever $\{x_n\}_{n=1}^\infty$ is a sequence in X converging to x_0 .

- (b) Let (X, d) and (Y, ρ) be metric spaces, and $f : X \rightarrow Y$ a uniformly continuous function. Show that the image of a Cauchy sequence in X under f is again a Cauchy sequence in Y . [6]